

Detailed VS Code & Dart Setup Guide

Step-by-Step Instructions with Visual Diagrams for Windows (Simple English)

This guide explains exactly how to set up your Windows computer for programming. Every tool is explained with small, clear actions so you cannot get lost.

Part 1: Install VS Code

VS Code is a clean, digital notebook program where you will write your code text.

- 1. Go to the Download Page:** Open your web browser (like Google Chrome), type `code.visualstudio.com` in the top address bar, and press Enter. Click the large blue button that says **"Download for Windows"**.
→ *Action:* Wait a minute for the setup installer file to completely download to your computer.
- 2. Open the Setup File:** Look at the bottom of your browser or go to your **Downloads** folder. Double-click the file named `VSCodeUserSetup.exe`.
→ *Action:* A window will open. Click the small circle next to **"I accept the agreement"** to consent to the terms, then click the **Next** button.
- 3. Click Through the Installer:** Keep clicking the **Next** button at the bottom of the window. You do not need to alter any settings or checkboxes.
→ *Action:* When you reach the screen with the **"Install"** button, click it. Watch the green progress bar load up.
- 4. Finish and Open:** Once the loading bar reaches the end, click the final **Finish** button. VS Code will pop open on your screen automatically!

Part 2: Download and Unzip Dart SDK

The **Dart SDK** is the core software engine that translates your text instructions into a language your computer understands.

- 1. Get the Safe Zip File:** Open your web browser and go to `dart.dev/get-dart`. Scroll down the page to the table grid and click the download button for Windows **x64 stable**. This downloads a compressed zip archive folder.
- 2. Extract (Unzip) the Folder:** Open your **Downloads** folder. Right-click on the newly downloaded Dart zip archive and select **"Extract All..."** from the menu.
→ *Action:* A prompt box will ask where to unzip it. Just click the **"Extract"** button at the bottom and wait for the loading bar to finish.
- 3. Move it to the Local Disk C:** Once extracted, you will see a regular folder called `dart - sdk`. Right-click this folder and select **"Copy"** (or click it and press `Ctrl + C`).

→ *Action:* On the left sidebar menu of your file screen, click **"This PC"**, double-click your **"Local Disk (C:)"**, right-click on any empty white space inside, and click **"Paste"** (or press `Ctrl + V`). Your path must be:

```
C:\dart-sdk
```

Part 3: Copy the Secret Bin Folder Path

This is the most important part. We must copy the exact digital location address of the core engine folder so Windows can find it.

1. **Open the dart-sdk Folder:** Go inside your **Local Disk (C:)**, locate the `dart-sdk` folder you just pasted, and double-click to open it.
2. **Enter the bin Folder:** Inside, you will see a folder called **bin**. Double-click to open it. This folder holds the main `dart.exe` application tool.
3. **Click the Top Address Bar:** Look at the very top of your file browser window. You will see a long, wide white path bar showing: *This PC > Local Disk (C:) > dart-sdk > bin*.
→ *Action:* Click your mouse once on the **empty white background area** at the far right side of that long bar.
4. **Copy the Path Text:** The folder text will immediately switch into a highlighted blue block of text that reads exactly: `C:\dart-sdk\bin`.
→ *Action:* Right-click that highlighted blue text and select **"Copy"** (or press `Ctrl + C` on your keyboard). Windows now has it stored safely in its memory!



Part 4: Setup Windows Environment Path Variable

We must paste our copied address into the system settings layout so your computer recognizes Dart anywhere.

1. **Open System Settings:** Press the **Windows Key** on your keyboard once, type `env` into the search bar, and click on the result named **"Edit the system environment variables"**.
2. **Open Environmental Menu:** A small grey menu box will appear. Click the button at the very bottom edge that says **"Environment Variables..."**.
3. **Find the Path Row:** Look at the lower grid area called *"System variables"*. Scroll down slightly to find the word row named **Path** (or **PATH**). Click it once so it lights up blue, then click the **"Edit..."** button.

4. **Paste the Address:** A new small list box will open. Click the **"New"** button on the right-hand side. A blank text line will appear at the bottom of the list.

→ *Action:* Right-click inside that blank line and select **"Paste"** (or press `Ctrl + V`). It should now show:

```
C:\dart-sdk\bin .
```

5. **Save and Close:** Click the **OK** button at the bottom of that box, click **OK** on the next box, and click **OK** on the last box. Make sure to click OK three times to lock in your work!

Part 5: Install Extensions in VS Code (Dart & Code Runner)

We need to add two extension plug-ins inside VS Code to help you write and instantly run your code scripts.

Plug-in 1: Dart Extension (For code grammar and syntax helper)

1. Open your ****VS Code**** desktop app.
2. Look at the far-left border menu strip. Click the ****Extensions**** icon button (it looks like four small box squares grouped up together).
3. Type `Dart` into the top search block. Find the first result named ****Dart**** (created by Dart Code) and click the blue ****Install**** button.

Plug-in 2: Code Runner Extension (Adds an instant "Run" button)

1. Click back into the top extensions search text field, clear out the text, and type `Code Runner`.
2. Look for the item named **Code Runner** (created by Jun Han). It features a distinct logo badge with a yellow run symbol.
3. Click the blue **Install** button next to it.
→ *Why you need it:* This extension adds a handy play button icon (▶) to the top right corner of VS Code. You can click it anytime to run your Dart files instantly without typing rules in the terminal!

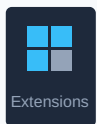


Diagram 2: Setting up both the Dart configuration helper and Code Runner execution widgets inside VS Code.

Part 6: Run a Quick Verification Test

Let's check if everything matches up perfectly by opening your Windows backend panel.

Hold down the `Windows Key` and tap the `R` key on your keyboard at the same time. This launches a tiny **"Run"** command prompt box on the bottom edge of your monitor screen.

2. Type `cmd` in the small white input line box and hit the Enter key. This loads up your black Command Prompt window panel.
3. Type this command line exactly as written and hit your Enter key: `dart --version`



Complete Success Check:

If you see text pop up showing `Dart SDK version 3.x.x` on the screen line, you are done! Your computer is completely ready to write code scripts and build apps.